

Romanization Flattens the Field: Tokenizer-Independent and Downstream Evidence of Script Fragility in Sinhala

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Abstract

Sinhala is written both in its native script and, in everyday digital use, romanized in Latin letters. A recent benchmark reported that language models are some 300 times worse on Romanized Sinhala and that model scale predicts nothing. We show both conclusions are properties of the metric. Recomputing 31 checkpoints with tokenizer-independent bits per byte, the perplexity and bits-per-byte rankings are uncorrelated ($\rho = 0.08$) because perplexity mostly tracks how finely a tokenizer shreds Sinhala, the 300-fold gap becomes 2.14 bits per byte, and scale does predict native-script ability. We then run the first downstream evaluation of Sinhala script variation, over three tasks, 9,479 items and ten instruction-tuned checkpoints in both scripts. Romanization does not degrade models uniformly, it flattens them: the loss is 0.78 times a checkpoint’s above-chance headroom ($R^2 = 0.97$), a relation holding again across 16 benchmark strata. Accuracy conceals this on offensive-language detection, where Matthews correlation loses 60% of its value.

1 Introduction

Sinhala has some 17 million speakers and a script of its own (Ranasinghe et al., 2025). Like several South Asian languages it has an everyday orthographic split: formal writing uses the native script, while chat and social media are dominated by Romanized Sinhala, typed in Latin letters with no standard spelling, so කොහොමද (“how are you”) appears as *kohomada*, *kohomd* or *kohomade*. This is not marginal. For Indian languages, Khullar et al. (2025) found 83% of Hindi and 73% of Telugu messages to a real health service were Romanized, and that models handled them worse. A Sinhala system is therefore likely to serve mostly Romanized input while being evaluated on native-script benchmarks.

Rajapakse and Weerasinghe (2026) gave the first systematic measurement, benchmarking 24 open-weight models with perplexity and reporting a median 312-fold degradation and no correlation between parameter count and script handling. Both rest on a metric that is not comparable across models, since perplexity is normalised by token count and token count depends entirely on the tokenizer (Gao et al., 2020; Magnusson et al., 2024), and Sinhala is where that matters because tokenizers segment it very unevenly (Petrov et al., 2023; Ahia et al., 2023). That work also notes perplexity says nothing about downstream ability, and calls for such evaluation. We address both problems.

Related work. Sinhala is low-resource with fragmented tooling (de Silva, 2019; Ranathunga and de Silva, 2022; Joshi et al., 2020) and its recent resources are almost entirely native-script (Pramodya et al., 2025; Ranasinghe et al., 2025; Dhananjaya et al., 2022; Chang et al., 2025). Romanized South Asian text has been studied mainly as a resource and identification problem (Roark et al., 2020; Benton et al., 2025), and J et al. (2024) show romanization can *help* when models are adapted for it, the mirror image of our zero-shot setting. Tokenizers treat languages unequally (Petrov et al., 2023; Ahia et al., 2023), which is why bits per byte exists (Gao et al., 2020; Magnusson et al., 2024). To our knowledge ours is the first work showing that, for a specific language, a published benchmark’s conclusions reverse when the normaliser changes.

C1: fixing the metric reverses the intrinsic conclusions. We recompute bits per byte, per character and per word for 31 checkpoints, 7 of them new, on the same parallel corpus (§3). Our perplexities reproduce the published values, so what follows comes from the metric, not the measurement. Perplexity and bits per byte are uncorrelated ($\rho = 0.08$, $p = 0.68$), while perplexity is almost

fully explained by tokenizer fertility ($\rho = -0.87$, $p < 10^{-9}$). The 300-fold gap is 2.14 bits per byte, and scale *does* predict native-script ability ($\rho = -0.55$, $p = 0.001$), though still not Romanized ability.

C2: Romanization flattens, by a measurable law. We evaluate ten instruction-tuned checkpoints on 9,479 items across multiple-choice knowledge, offensive-language detection and physical commonsense, in both scripts, for 189,580 generations (§4). Native-script ability varies widely and Romanized ability barely does: the loss is $0.78\times$ a checkpoint’s above-chance headroom ($R^2 = 0.97$), and the same relation reappears across the 16 domain by difficulty cells. The effect is neither a token-length artifact nor degeneration.

C3: what predicts fragility, and released artifacts. Absolute native-script bits per byte predicts the downstream gap ($\rho = -0.90$, $p = 0.0009$), whereas the bits-per-byte *gap*, the quantity prior work reports, predicts nothing ($\rho = -0.07$, $p = 0.86$). Building the Romanized side needed a native-to-Latin transliterator, the opposite of the direction most Sinhala tools support; ours beats three alternatives on 755,000 human references and is released with the paired evaluation sets and all per-item generations (Appendix A).

2 Setup

Conditions and data. Every item is scored twice, in the native script as published and romanized with identical content, so all comparisons are paired and differ only in orthography. The intrinsic corpus is the parallel set of [Rajapakse and Weerasinghe \(2026\)](#): 500 sentence pairs whose Romanized side was *typed by native speakers*, so those results do not depend on our transliterator, plus 500 authentic mixed-script sentences. Downstream we use every available item, with no sampling. SinhalaMMLU ([Pramodya et al., 2025](#)) is 6,879 curriculum-aligned multiple-choice items over six domains and three difficulty levels; 2,139 have five options, so guessing scores 23.4%, not 25%.¹ SOLD ([Ranasinghe et al., 2025](#)) is the full 2,500-item offensive-language test split, 40.6% of-fen-sive, so the majority class scores 59.4%. Global PIQA ([de Silva and Ranathunga, 2026](#)) adds 100 Sinhala items.

¹The public release is an 1,851-item sample. We obtained the full set from the authors.

Transliteration. Almost all Sinhala tools run Latin to native ([De Mel et al., 2025](#); [Sumanathilaka et al., 2025](#)), so we wrote the reverse: a deterministic grapheme-level transliterator using the conventions people type, *ඞ* as *th* and *ඞ* as *w* not *v*, so *වෙසක් උත්සවයේදී* becomes *wesak uthsaway-eedii*. Against 755,000 human-romanized items ([Sumanathilaka et al., 2025](#)) it has the lowest character error rate of four methods on all three reference corpora, beating Aksharamukha ([Rajan](#)) and uroman ([Hermjakob et al., 2018](#)) at $p < 10^{-90}$ (Appendix E).

Models and prompting. The intrinsic evaluation covers 31 decoder-only checkpoints from 350M to 14.7B across eight families (Appendix I). Downstream needs models that follow a response format, so it uses the ten instruction-tuned checkpoints in the pool, 1.1B to 9B. All prompts are zero-shot and greedily decoded with a 40-token limit. One template, chosen by a 3,240-generation pilot, is applied to every dataset and both conditions; what matters for a paired comparison is not that the template be optimal but that it not favour a script, and the pilot winner was the same under both scripts everywhere (Appendix B). Un-parseable answers are scored wrong and reported as an invalid rate.

Statistics and competence screen. Since both conditions score the same items we use McNemar’s test ([McNemar, 1947](#); [Dror et al., 2018](#)), report discordant pair counts, and control the family-wise error rate over checkpoints within each dataset with Holm’s procedure ([Holm, 1979](#)). Intervals come from 10,000 paired item bootstrap resamples. For SOLD we lead with Matthews correlation ([Matthews, 1975](#); [Chicco and Jurman, 2020](#)), since the classes are imbalanced and accuracy sits near the majority baseline. A model at chance natively cannot lose anything by being given Romanized input, so a small gap for it is evidence of nothing and averaging it in understates the effect. We therefore mark a checkpoint *competent* on a task when its native score beats the guessing baseline at $p < 0.01$ under a Poisson-binomial null and it puts at most 90% of answers on one option. Five of ten pass on SinhalaMMLU, five have a non-trivial SOLD signal, none pass on Global PIQA, and all ten appear in every table.

3 Perplexity Measures Tokenizers

We divide the total negative log-likelihood in nats by a unit of the text itself rather than by its tokenization, $\text{BPB} = \sum_s \text{NLL}(s) / (\ln 2 \sum_s |s|_{\text{bytes}})$, and likewise per character (BPC) and per whitespace word (BPW), pooling numerator and denominator over the corpus so short sentences do not dominate. Bytes are standard (Gao et al., 2020), but BPW is sharpest across scripts, since a sentence and its romanization contain the same number of words while Sinhala UTF-8 spends 2.65 bytes per character against 1.00 for Latin. Our perplexities reproduce the published values for all 24 shared checkpoints, median absolute deviation 0.05% native and 0.004% Romanized.

The two metrics disagree. Figure 1(a) is the methodological result. Native-script perplexity and bits per byte are statistically unrelated ($\rho = 0.08$, $p = 0.68$; Romanized $\rho = 0.13$, $p = 0.48$), only one of the five best checkpoints by perplexity is among the five best by bits per byte, and Gemma-2-9B ranks 26th of 31 on perplexity but 2nd on bits per byte. The cause is the colour axis: perplexity is almost fully explained by tokens spent per Sinhala word ($\rho = -0.87$, $p < 10^{-9}$), from 4.3 for the Qwen3.5 and Gemma tokenizers to 14.2 for models falling back to bytes. Spreading the same loss over three times as many tokens lowers per-token perplexity without the model knowing more, so a cross-family perplexity comparison here is largely a comparison of tokenizers.

Two published conclusions do not survive. The 312-fold degradation is mostly tokenization: our median ratio is 294-fold, but the same gap is 2.14 bits per byte and 1.11 bits per word. And scale is uncorrelated with native perplexity ($\rho = 0.15$, $p = 0.42$) yet correlated with native bits per byte ($\rho = -0.55$, $p = 0.001$). Scale does buy native-script ability; what it does not buy is Romanized ability ($\rho = -0.28$, $p = 0.13$).

Flattening, and an absolute reference. Native-script cost spans 16.3 to 34.1 bits per word across checkpoints while Romanized cost spans only 20.3 to 25.7, so a full bit per word of native improvement buys 0.14 bits on Romanized text (95% CI [0.05, 0.25]), and 12 of 31 checkpoints are actually *cheaper* on Romanized than on native Sinhala: the better a checkpoint is natively, the more it loses ($\rho = -0.88$, Appendix C). Mixed script instead reproduces the earlier finding, rank-correlated with

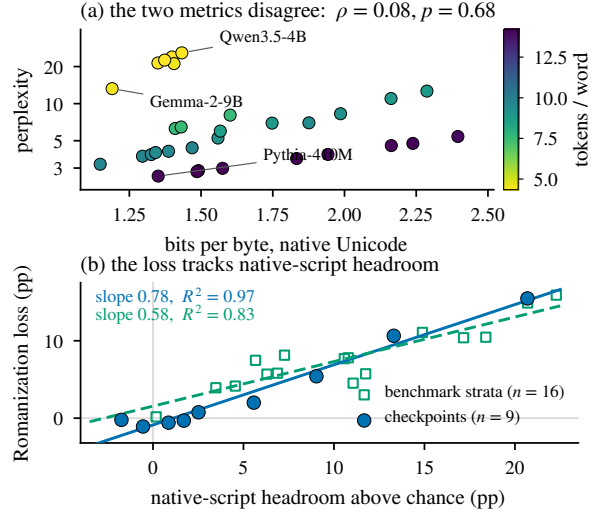


Figure 1: (a) Native-script perplexity is unrelated to native-script bits per byte across 31 checkpoints, and is instead determined by tokenizer fertility (colour). (b) Romanization loss against native-script headroom above chance, measured independently across checkpoints and across the 16 domain by difficulty cells of SinhalaMMLU.

native at $\rho = 0.94$, so interleaving scripts is a mild perturbation where replacing one is not. To calibrate absolutely, add- α character n -gram models fit on the same corpus under five-fold cross-validation reach 2.64 bits per byte on Romanized Sinhala and a character bigram reaches 3.06, while the best of 31 checkpoints reaches 3.29 and the median 3.81: not one beats the bigram, though on native Sinhala the trigram is at 1.16 with one checkpoint below it (Appendix D). Whatever these checkpoints know about Sinhala, they know it about the script rather than the language.

4 Downstream Consequences

Multiple-choice knowledge. Table 1 reports every checkpoint. Four of the five competent checkpoints lose significantly after Holm correction: Qwen3.5-9B falls from 44.1% to 28.7%, a 15.5 point drop (95% CI [14.0, 16.9], 1,892 against 828 discordant pairs), Qwen3.5-4B loses 10.6, Llama-3.1-8B-Instruct 5.4 and Qwen2-7B-Instruct 2.0. The fifth, Hormoz-8B, starts only 2.5 points above chance and loses 0.8, not significant. Pooled the gap is 6.9 points (95% CI [6.3, 7.5], $n = 34,395$ paired items), and a logistic fit with checkpoint fixed effects and item-clustered errors puts the odds ratio for Romanized input at 0.68 ($p < 10^{-30}$). Read as levels this is starker: native accuracy spans 22.4 points and Romanized only 6.7,

Checkpoint	SinhalaMMLU (6,879 items)						SOLD (2,500 items)				Invalid %			
	Acc. %					p_{Holm}	macro-F1		MCC		p_{Holm}	Invalid %		
	B	U	R	Δ	95% CI		U	R	U	R		U	R	τ
TinyLlama-1.1B-Chat	1.1	21.7	21.9	-0.2	[-0.5,0.1]	.8273	6.4	24.2	-0.019	-0.043	1.0000	1.5	1.0	0.45
LaMini-GPT-1.5B	1.5	0.9	18.8	-17.9	[-18.8,-16.9]	<.0001	5.1	28.3	0.000	0.024	1.0000	95.9	16.4	0.31
StableLM-Zephyr-3B	2.8	22.9	24.0	-1.1	[-2.3,0.1]	.4275	38.6	38.9	0.036	0.001	.9288	3.8	2.6	0.32
SmolLM3-3B	3.1	25.1	25.5	-0.3	[-1.0,0.4]	.8273	29.7	29.9	0.025	0.017	1.0000	0.0	0.0	0.48
Qwen3.5-4B†	4.0	36.7	26.1	+10.6	[9.3,12.0]	<.0001	60.2	55.1	0.219	0.127	.0014	0.0	0.0	0.72
Zephyr-7B-beta	7.2	24.3	24.9	-0.6	[-1.7,0.5]	.8273	39.7	39.6	0.060	0.060	1.0000	2.1	0.6	0.49
Qwen2-7B-Instruct†	7.6	29.0	27.0	+2.0	[0.8,3.2]	.0047	49.9	50.3	0.148	0.042	<.0001	0.0	0.0	0.51
Llama-3.1-8B-Instruct†	8.0	32.5	27.1	+5.4	[4.1,6.7]	<.0001	60.1	52.6	0.259	0.105	<.0001	0.0	0.0	0.45
Hormoz-8B†	8.0	26.0	25.2	+0.8	[-0.5,2.0]	.8273	47.9	51.9	0.108	0.090	1.0000	0.0	0.0	0.50
Qwen3.5-9B†	9.0	44.1	28.7	+15.5	[14.0,16.9]	<.0001	63.1	52.0	0.280	0.069	<.0001	0.0	0.0	0.72

Table 1: Downstream results by size. U is native Unicode, R Romanized, Δ the paired difference in points, p_{Holm} McNemar’s test corrected across checkpoints within a task (mcc: paired bootstrap). τ is Romanized over native input tokens on SinhalaMMLU, below one throughout. Invalid % is unparseable generations. † passes the competence screen (§2).

never exceeding 28.7% against a 23.4% baseline, so the best checkpoint keeps 5.2 of its 20.7 points of headroom, and the six checkpoints showing no gap are the six already at chance.

You can only lose what you have. That relation is tight. Writing headroom as native accuracy minus chance, a fit across the nine parseable checkpoints gives

$$\text{loss} = 0.78 \times \text{headroom} - 0.87, \quad (1)$$

with $R^2 = 0.97$ and $p = 1.9 \times 10^{-6}$ (Figure 1b): Romanization removes roughly four fifths of whatever native ability a checkpoint had. A regression on nine points is thin, so we test the same relation inside the benchmark, where the units are items rather than models: over the 16 non-empty domain by difficulty cells, headroom and loss per cell give a slope of 0.58 with $R^2 = 0.83$ ($p = 9.8 \times 10^{-7}$). The two decompositions are independent, one across models and one across content, and they agree. Loss also falls monotonically with difficulty, 8.6 points on Easy to 5.2 on Hard ($p = 0.007$ for the interaction), because Hard items sit nearer chance in both conditions (Appendix F): the harder a slice, the smaller the script effect it can reveal.

Accuracy hides the damage on SOLD. Accuracy falls by only 1.6 to 11.3 points against a 59.4% majority baseline, so three checkpoints look almost unaffected. Matthews correlation says otherwise: Qwen3.5-9B goes from 0.280 to 0.069, keeping 24% of its signal, Qwen2-7B-Instruct from 0.148 to 0.042 (29%) and Llama-3.1-8B-Instruct from 0.259 to 0.105 (40%). Four of the five checkpoints

with a real native signal lose significantly under Holm correction and the median keeps 40%. The mechanism is a retreat toward trivial behaviour: Qwen2-7B-Instruct more than doubles its rate of predicting the offensive class, 10.5% to 23.4%, without becoming more accurate. A deployment tracking accuracy would see a 5 point wobble while over half the usable signal disappeared.

Two things it is not. The natural explanation is that romanization shatters words into more tokens. For Sinhala the opposite holds: the native script costs a median 9.5 tokens per word against 2.09 for its romanization, because these tokenizers largely fall back to bytes, and all ten checkpoints receive *fewer* tokens in the Romanized condition on all three tasks, a median ratio of 0.48 (column τ). Models get shorter, more compactly encoded input and do worse, and terciles by token inflation show no monotone pattern either. Nor is this degeneration: the most frequent option takes 37.0% of answers natively and 38.6% romanized (Wilcoxon $p = 0.63$) and the invalid rate never exceeds 0.03% for competent checkpoints, so models keep answering and are simply wrong. Global PIQA is a null result (Appendix H).

What this means in practice. Perplexity reported across model families on a script tokenizers handle unevenly measures tokenizer behaviour, which for Sinhala inverts both the ranking and the conclusion about scale. Script robustness also cannot be read off a group average, since six of our ten checkpoints contribute a zero gap by construction, so such claims need a competence screen and a stated baseline.

Limitations

One language. Everything here is Sinhala. The mechanism, a floor effect making robustness gaps proportional to native ability, should generalise to any language with a dominant Romanized register, but we have not shown that. Hindi, Urdu, Nepali, Tamil and Arabic dialects are the obvious tests, and [Khullar et al. \(2025\)](#) report compatible magnitudes for the first three.

The Romanized downstream side is synthetic. We transliterate native-script items rather than collecting human-typed Romanized versions, because no parallel resource exists at this scale. Our transliterator is the best of four on 755,000 human references, and its residual error is a spelling regularity that should make the text easier rather than harder, but it is not what a person would type. The intrinsic results do not share this limitation, since their Romanized side was typed by native speakers, and they show the same flattening. A human-written Romanized version of even a few hundred Sinhala-MMLU items would settle the question.

Open weights only, and small ones. The downstream evaluation covers open checkpoints up to 9B. Frontier proprietary models are much stronger on Sinhala ([Pramodya et al., 2025](#)) and so have far more headroom, which means Equation 1 predicts larger absolute losses for them. That prediction is untested here.

Zero-shot only. One template, zero-shot, greedy. This measures what a user gets out of the box, not the ceiling. Few-shot prompting, transliterate-then-infer, or Romanized fine-tuning could all narrow the gap and we do not measure by how much.

Two checkpoints cannot do the task. LaMini-GPT-1.5B and TinyLlama-1.1B-Chat produce unparseable output on most native-script items of at least one task, so their apparent improvement under Romanization is an artifact of that failure rather than a script effect. This is why the competence screen exists.

The n -gram comparison is not matched. The n -gram models are fit in-domain on held-out folds of the same corpus while the checkpoints are zero-shot. The comparison establishes an absolute floor, not a like-for-like ranking.

Ethics Statement

All datasets are public research releases used within their licences and for evaluation only, which matters for Global PIQA, whose licence forbids training. SOLD contains offensive social media text, already pseudonymised upstream with user mentions replaced by placeholders; we neither collected nor re-annotated any of it, and we add no new human annotation. The practical risk our results speak to is deployment: a moderation or support system validated on native-script Sinhala and serving Romanized users will look far better in evaluation than in the field, and on SOLD that failure is invisible in accuracy while more than half the usable signal is gone. Our transliterator collapses several Sinhala consonant distinctions, matching casual practice, so it is a tool for building evaluation data rather than for rendering names or other text where those distinctions carry meaning.

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References

- Marah Abdin, Jyoti Aneja, Harkirat Behl, Sébastien Bubeck, Ronen Eldan, Suriya Gunasekar, et al. 2024. [Phi-4 technical report](#). *Preprint*, arXiv:2412.08905.
- Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David Mortensen, Noah Smith, and Yulia Tsvetkov. 2023. [Do all languages cost the same? tokenization in the era of commercial language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9904–9923, Singapore. Association for Computational Linguistics.
- Adrian Benton, Alexander Gutkin, Christo Kirov, and Brian Roark. 2025. [Improving informally Romanized language identification](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 2318–2336, Suzhou, China. Association for Computational Linguistics.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, et al. 2023. [Pythia: A suite for analyzing large language models across training and scaling](#). *Preprint*, arXiv:2304.01373.
- BigScience Workshop, Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, et al. 2023. [BLOOM: A 176B-parameter open-access multilingual language model](#). *Preprint*, arXiv:2211.05100.

- Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. 2020. [PIQA: Reasoning about physical commonsense in natural language](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 7432–7439.
- Tyler A. Chang, Catherine Arnett, Abdelrahman Sadallah, Abdelrahman Eldesokey, et al. 2025. [Global PIQA: Evaluating commonsense reasoning across 100+ languages and cultures](#). *Preprint*, arXiv:2510.24081.
- Davide Chicco and Giuseppe Jurman. 2020. [The advantages of the Matthews correlation coefficient \(MCC\) over F1 score and accuracy in binary classification evaluation](#). *BMC Genomics*, 21:6.
- Yomal De Mel, Kasun Wickramasinghe, Nisansa de Silva, and Surangika Ranathunga. 2025. [Sinhala transliteration: A comparative analysis between rule-based and Seq2Seq approaches](#). In *Proceedings of the First Workshop on Natural Language Processing for Indo-Aryan and Dravidian Languages*, pages 166–173, Abu Dhabi. Association for Computational Linguistics.
- Nisansa de Silva. 2019. [Survey on publicly available Sinhala natural language processing tools and research](#). *Preprint*, arXiv:1906.02358.
- Nisansa de Silva and Surangika Ranathunga. 2026. [Sinhala physical common sense reasoning dataset for global PIQA](#). *Preprint*, arXiv:2602.02207.
- Vinura Dhananjaya, Piyumal Demotte, Surangika Ranathunga, and Sanath Jayasena. 2022. [BERTifying Sinhala: A comprehensive analysis of pre-trained language models for Sinhala text classification](#). In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 7377–7385, Marseille, France. European Language Resources Association.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. [The hitchhiker’s guide to testing statistical significance in natural language processing](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392, Melbourne, Australia. Association for Computational Linguistics.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. 2020. [The Pile: An 800GB dataset of diverse text for language modeling](#). *Preprint*, arXiv:2101.00027.
- Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhatipatiraju, et al. 2024. [Gemma 2: Improving open language models at a practical size](#). *Preprint*, arXiv:2408.00118.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, et al. 2024. [The Llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. [Measuring massive multitask language understanding](#). In *International Conference on Learning Representations (ICLR)*.
- Ulf Hermjakob, Jonathan May, and Kevin Knight. 2018. [Out-of-the-box universal Romanization tool uroman](#). In *Proceedings of ACL 2018, System Demonstrations*, pages 13–18, Melbourne, Australia. Association for Computational Linguistics.
- Sture Holm. 1979. [A simple sequentially rejective multiple test procedure](#). *Scandinavian Journal of Statistics*, 6(2):65–70.
- Jaavid J, Raj Dabre, Aswanth M, Jay Gala, Thanmay Jayakumar, Ratish Puduppully, and Anoop Kunchukuttan. 2024. [RomanSetu: Efficiently unlocking multilingual capabilities of large language models via Romanization](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15593–15615, Bangkok, Thailand. Association for Computational Linguistics.
- Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. [The state and fate of linguistic diversity and inclusion in the NLP world](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293, Online. Association for Computational Linguistics.
- Manurag Khullar, Utkarsh Desai, Poorva Malviya, Aman Dalmia, and Zheyuan Ryan Shi. 2025. [Script gap: Evaluating LLM triage on Indian languages in native vs romanized scripts in a real world setting](#). *Preprint*, arXiv:2512.10780.
- Hugo Laurençon, Lucile Saulnier, Thomas Wang, Christopher Akiki, Albert Villanova del Moral, et al. 2023. [The BigScience ROOTS corpus: A 1.6TB composite multilingual dataset](#). *Preprint*, arXiv:2303.03915.
- Ian Magnusson, Akshita Bhagia, Valentin Hofmann, Luca Soldaini, Ananya Harsh Jha, Oyvind Tafjord, Dustin Schwenk, Evan Pete Walsh, Yanai Elazar, Kyle Lo, Dirk Groeneveld, Iz Beltagy, Hannaneh Hajishirzi, Noah A. Smith, Kyle Richardson, and Jesse Dodge. 2024. [Paloma: A benchmark for evaluating language model fit](#). In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*.
- B. W. Matthews. 1975. [Comparison of the predicted and observed secondary structure of T4 phage lysozyme](#). *Biochimica et Biophysica Acta (BBA) – Protein Structure*, 405(2):442–451.

Quinn McNemar. 1947. [Note on the sampling error of the difference between correlated proportions or percentages](#). *Psychometrika*, 12(2):153–157.

Aleksandar Petrov, Emanuele La Malfa, Philip Torr, and Adel Bibi. 2023. [Language model tokenizers introduce unfairness between languages](#). In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Maja Popović. 2015. [chrF: character n-gram F-score for automatic MT evaluation](#). In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, Lisbon, Portugal. Association for Computational Linguistics.

Ashmari Pramodya, Nirasha Nelki, Heshan Shalinda, Chamila Liyanage, Yusuke Sakai, Randil Pushpananda, Ruvan Weerasinghe, Hidetaka Kamigaito, and Taro Watanabe. 2025. [SinhalaMMLU: A comprehensive benchmark for evaluating multitask language understanding in Sinhala](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 32943–32961, Suzhou, China. Association for Computational Linguistics.

Vinodh Rajan. [Aksharamukha: Script converter for indic and southeast asian scripts](#). Software. Version 2.3, accessed August 2026.

Minuri Rajapakse and Ruvan Weerasinghe. 2026. [Script sensitivity: Benchmarking language models on Unicode, romanized and mixed-script Sinhala](#). In *2026 9th IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE)*, pages 1–6. IEEE.

Tharindu Ranasinghe, Isuri Anuradha, Damith Premasiri, Kanishka Silva, Hansi Hettiarachchi, Lasitha Uyangodage, and Marcos Zampieri. 2025. [SOLD: Sinhala offensive language dataset](#). *Language Resources and Evaluation*, 59:297–337.

Surangika Ranathunga and Nisansa de Silva. 2022. [Some languages are more equal than others: Probing deeper into the linguistic disparity in the NLP world](#). In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 823–848, Online only. Association for Computational Linguistics.

Brian Roark, Lawrence Wolf-Sonkin, Christo Kirov, Sabrina J. Mielke, Cibu Johny, Isin Demirsahin, and Keith Hall. 2020. [Processing South Asian languages written in the Latin script: the Dakshina dataset](#). In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 2413–2423, Marseille, France. European Language Resources Association.

Deshan Sumanathilaka, Sameera Perera, Sachithya Dharmasiri, Maneesha Athukorala, Aloka D. Herath,

Rukshan Dias, Piyumal Gamage, Ruvan Weerasinghe, and Y. H. P. Priyadarshana. 2025. [Swabhasha resource hub: Romanized Sinhala to Sinhala transliteration systems and data resources](#). *Preprint*, arXiv:2507.09245.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, et al. 2025. [Qwen3 technical report](#). *Preprint*, arXiv:2505.09388.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, et al. 2024. [Qwen2 technical report](#). *Preprint*, arXiv:2407.10671.

A Released Artifacts and Reproducibility

We release three artifacts at <https://anonymous.4open.science/r/sinhala-script-robustness>. The **transliterater** (`pip install sinlipi`, <https://pypi.org/project/sinlipi>) is local, deterministic, covers every input and is the best of four methods on 755,000 human references, which makes Romanized evaluation data cheap to build for any native-script Sinhala resource. The **frozen evaluation sets** pair both script conditions in one record per item, regenerable byte-identically with SHA-256 digests recorded, so a paired analysis cannot compare mismatched subsets. The **harness and analysis pipeline** ships all 189,580 per-item generations, the statistics, the figure and table generators, and a checker that restates every number in this paper and verifies it against the analysis output, so the reported values cannot drift from the data.

B Prompt Template and Compute

SinhalaMMLU follows MMLU (Hendrycks et al., 2021) and Global PIQA follows PIQA (Bisk et al., 2020). One template is used for every dataset and both script conditions. Multiple choice, with the option count n and the domain name inserted where available:

```
This is a multiple-choice
question related to the
{domain}. Choose the correct
or most appropriate answer
from answers 1/.../n for the
following question. Do not
repeat the question or text
in your response. Respond with
only the number (1/.../n) and
nothing else.
Question: {question}
1. {option 1} ... n. {option
n}
Answer:
```

Binary classification:

Classify the following Sinhala text as either 'NOT' (not offensive) or 'OFF' (offensive). Do not repeat the question or text in your response. Respond with only 'NOT' or 'OFF' and nothing else.
Text: {text}
Answer:

Prompts are rendered through each checkpoint’s own chat template. For checkpoints that emit reasoning traces by default we disable that behaviour identically in both script conditions. Decoding is greedy at temperature 0 with at most 40 new tokens and inputs truncated at 4,096 tokens. Models run in fp16 on dual T4 GPUs. Intrinsic evaluation scores 46,500 sequences (31 checkpoints, 1,500 each); downstream evaluation produces 189,580 generations (10 checkpoints, 9,479 items, two conditions).

C Flattening

D Character n -gram Reference

Model	Native Unicode		Romanized	
	BPB	#LMs <	BPB	#LMs <
character 1-gram	1.815	22	3.977	24
character 2-gram	1.352	7	3.063	0
character 3-gram	1.162	1	2.685	0
character 4-gram	1.269	2	2.636	0
character 5-gram	1.474	15	2.947	0
character 6-gram	1.680	21	3.381	3
best checkpoint	1.148	–	3.290	–
median checkpoint	1.486	–	3.805	–

Table 2: Held-out character n -gram models against the 31 checkpoints. “#LMs <” counts checkpoints reaching lower bits per byte than the n -gram model. No checkpoint beats a character bigram on Romanized Sinhala.

E Transliteration Evaluation

Each hypothesis is scored against the closest accepted human variant, case-folded, using character error rate, chrF (Popović, 2015) and exact match. Case is folded because the web tool capitalises the first letter of whatever it is given, an interface artifact rather than a romanization choice, and scoring case-sensitively would reward it for that alone. The web tool is close once its v is rewritten to w , a spelling convention rather than a transliteration error, but it cannot romanize 17 sequences involving ඳ් at all and leaves Sinhala characters in 0.8%

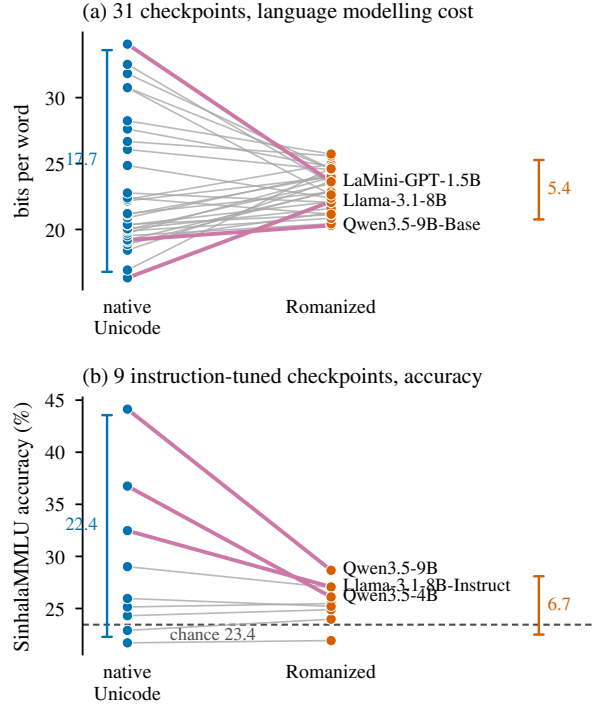


Figure 2: Romanization flattens the field. (a) Language modelling cost in bits per word for 31 checkpoints; word counts are identical within a pair, so the two columns are directly comparable. Native-script cost varies by 17.7 bits across checkpoints, Romanized cost by 5.4, and crossing lines mark the twelve checkpoints cheaper on Romanized text. (b) SinhalaMMLU accuracy. A 22.4 point spread becomes 6.7, and no checkpoint finishes more than 5.3 points above chance. Brackets give each column’s spread. LaMini-GPT-1.5B is omitted from (b) because it produces unparseable output on 95.9% of native-script items; it appears in Table 1.

to 2.5% of its output. All four methods double long vowels far more often than people do, 0.81 per word against 0.12, which is the dominant residual error; canonicalising that single convention on both sides cuts our character error rate to 0.040 on the word corpus.

Method	CER ↓			chrF	Exact	Cov.
	sent.	word	aug.	sent.	word	%
Ours	0.182	0.121	0.112	67.8	33.2	100.00
Web tool $v \rightarrow w$	0.182	0.133	0.114	67.8	30.9	99.67
Aksharamukha	0.191	0.148	0.139	63.7	23.9	100.00
Web tool, raw	0.197	0.176	0.153	63.4	20.6	99.67
uroman	0.228	0.228	0.210	55.7	9.7	100.00

Table 3: Transliteration quality against human references: 4,397 authentic social-media sentence pairs, 450,587 multi-reference words, and a fixed 300,000-sentence sample of a machine-augmented set used only as a cross-check. Coverage is the share of inputs the method produced any output for.

F SinhalaMMLU Strata

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Stratum	Items	U %	R %	Δ	p_{Holm}
<i>by difficulty</i>					
Easy	1,851	37.7	29.1	8.6	< .0001
Medium	2,503	35.9	28.7	7.2	< .0001
Hard	2,525	28.5	23.2	5.2	< .0001
<i>by domain</i>					
Social Science	1,059	38.5	26.7	11.8	< .0001
Business Studies	463	35.2	27.1	8.0	< .0001
Stem	614	34.2	26.2	8.0	< .0001
Other	1,014	32.5	24.5	8.0	< .0001
Humanities	3,341	32.7	27.9	4.8	< .0001
Language	388	29.4	24.7	4.7	.0002

Table 4: SinhalaMMLU by stratum, pooled over the five competent checkpoints. Item counts are per checkpoint. Domains are ordered by loss.

G Intrinsic to Downstream Linkage

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Intrinsic predictor	MMLU gap		SOLD Δ_{MCC}	
	ρ	p	ρ	p
Unicode BPB (level)	-0.90	.0009	-0.82	.0072
Romanized BPB (level)	-0.85	.0037	-0.92	.0005
BPB gap (Rom. - Uni.)	-0.07	.8647	-0.40	.2861
Unicode BPW (level)	-0.90	.0009	-0.82	.0072
perplexity ratio	-0.53	.1392	-0.42	.2646
parameter count	+0.65	.0567	+0.49	.1768

Table 5: Spearman correlations between intrinsic measurements and downstream script effects over the nine parseable checkpoints. Levels predict; gaps and ratios do not.

H Global PIQA

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Checkpoint	Acc. %			disc. b/c	p_{Holm}	Inv. %	
	U	R	Δ			U	R
TinyLlama-1.1B-Chat	40	33	+7	12/5	1.0000	18	28
LaMini-GPT-1.5B	0	36	-36	0/36	< .0001	100	28
StableLM-Zephyr-3B	45	44	+1	15/14	1.0000	0	0
SmolLM3-3B	53	51	+2	14/12	1.0000	0	0
Qwen3.5-4B	57	49	+8	22/14	1.0000	0	0
Zephyr-7B-beta	48	50	-2	18/20	1.0000	0	0
Qwen2-7B-Instruct	45	42	+3	21/18	1.0000	0	0
Llama-3.1-8B-Instruct	53	48	+5	22/17	1.0000	0	0
Hormoz-8B	47	48	-1	13/14	1.0000	0	0
Qwen3.5-9B	53	45	+8	25/17	1.0000	0	0

Table 6: Global PIQA, 100 items. “disc. b/c” are the discordant pair counts entering McNemar’s test.

I Checkpoints and Full Intrinsic Results

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Checkpoint	Native Unicode				Romanized			Mixed script			Hugging Face identifier
	B	PPL	BPB	BPW	PPL	BPB	BPW	PPL	BPB	BPW	
Llama-3.1-8B	8.0	3.22	1.148	16.33	970.5	3.584	22.12	6.36	1.641	19.66	meta-llama/Meta-Llama-3.1-8B
Gemma-2-9B	9.2	13.23	1.190	16.92	4726.1	3.999	24.68	48.99	1.792	21.47	google/gemma-2-9b
Llama-3.2-3B	3.2	3.74	1.296	18.42	1560.7	3.831	23.65	7.40	1.776	21.27	meta-llama/Llama-3.2-3B
Llama-3.1-8B-Instruct‡	8.0	3.86	1.326	18.86	1255.5	3.718	22.95	7.94	1.839	22.03	meta-llama/Llama-3.1-8B-Instruct
Phi-4-14B	14.7	4.01	1.341	19.07	1149.1	3.446	21.27	8.08	1.832	21.94	microsoft/phi-4
Qwen3.5-9B-Base	9.0	21.39	1.350	19.19	1250.1	3.290	20.30	78.56	1.958	23.46	Qwen/Qwen3.5-9B-Base
Pythia-410M	0.4	2.58	1.351	19.20	1245.0	3.707	22.88	4.91	1.978	23.69	EleutherAI/pythia-410m
Qwen3.5-9B‡	9.0	22.56	1.373	19.52	1307.8	3.311	20.43	84.24	1.990	23.83	Qwen/Qwen3.5-9B
Llama-3.2-1B	1.2	4.11	1.387	19.72	1718.6	3.881	23.96	8.54	1.903	22.79	meta-llama/Llama-3.2-1B
Qwen3.5-4B-Base	4.0	23.92	1.399	19.89	1512.7	3.378	20.85	83.32	1.985	23.77	Qwen/Qwen3.5-4B-Base
Gemma-7B	8.5	21.13	1.406	19.99	3138.9	3.805	23.48	66.00	1.930	23.11	google/gemma-7b
Qwen2-7B	7.6	6.29	1.410	20.05	1357.0	3.498	21.59	14.11	1.919	22.98	Qwen/Qwen2-7B
Qwen2-7B-Instruct‡	7.6	6.46	1.431	20.34	1611.2	3.581	22.10	15.45	1.985	23.77	Qwen/Qwen2-7B-Instruct
Qwen3.5-4B‡	4.0	25.84	1.433	20.37	1663.1	3.422	21.12	90.71	2.023	24.23	Qwen/Qwen3.5-4B
TinyLlama-1.1B-Chat‡	1.1	4.38	1.469	20.89	1014.2	3.923	24.21	8.92	1.980	23.71	TinyLlama/TinyLlama-1.1B-Chat-v1.0
Mistral-Nemo-Base-2407	12.2	2.80	1.486	21.12	1938.4	3.896	24.04	4.63	1.912	22.90	mistralai/Mistral-Nemo-Base-2407
Cerebras-GPT-1.3B	1.3	2.86	1.491	21.20	1040.2	3.700	22.83	5.23	2.052	24.58	cerebras/Cerebras-GPT-1.3B
Mistral-7B-v0.3	7.2	5.26	1.559	22.17	921.4	3.884	23.97	10.15	1.984	23.76	mistralai/Mistral-7B-v0.3
Hormoz-8B‡	8.0	5.98	1.567	22.28	4175.8	4.045	24.96	13.58	2.083	24.94	mann-e/Hormoz-8B
Minitron-8B-Base	8.4	2.98	1.575	22.39	2118.6	3.428	21.15	5.08	2.000	23.95	nvidia/Minitron-8B-Base
Qwen1.5-1.8B	1.8	8.07	1.601	22.77	1584.9	3.573	22.05	17.60	2.079	24.90	Qwen/Qwen1.5-1.8B
BLOOM-3B	3.0	6.95	1.748	24.85	3258.5	3.626	22.38	15.90	2.254	26.99	bigscience/bloom-3b
OPT-2.7B	2.7	3.59	1.833	26.06	1052.8	3.974	24.53	6.26	2.300	27.54	facebook/opt-2.7b
SmolLM3-3B‡	3.1	6.98	1.876	26.67	5373.0	4.145	25.58	13.89	2.301	27.56	HuggingFaceTB/SmolLM3-3B
OPT-1.3B	1.3	3.87	1.942	27.62	1121.9	4.010	24.75	6.82	2.407	28.83	facebook/opt-1.3b
Zephyr-7B-beta‡	7.2	8.28	1.986	28.24	1515.6	4.167	25.72	15.76	2.360	28.27	HuggingFaceH4/zephyr-7b-beta
BLOOM-1B1	1.1	11.00	2.162	30.74	3558.5	3.666	22.62	24.52	2.607	31.22	bigscience/bloom-1b1
StableLM-Zephyr-3B‡	2.8	4.57	2.163	30.76	2129.2	3.986	24.60	7.51	2.509	30.05	stabilityai/stablelm-zephyr-3b
OPT-350M	0.3	4.75	2.238	31.82	1075.9	3.986	24.60	8.15	2.630	31.50	facebook/opt-350m
BLOOM-560M	0.6	12.64	2.287	32.52	4915.0	3.810	23.52	28.01	2.715	32.52	bigscience/bloom-560m
LaMini-GPT-1.5B‡	1.5	5.41	2.395	34.06	1322.0	3.827	23.62	9.46	2.788	33.39	MBZUAI/LaMini-GPT-1.5B

Table 7: Corpus-pooled intrinsic metrics for all 31 checkpoints, sorted by native-script bits per byte, with Hugging Face identifiers. The pool spans the Llama (Grattafiori et al., 2024), Qwen (Yang et al., 2024, 2025), Gemma (Gemma Team et al., 2024), Mistral, BLOOM (BigScience Workshop et al., 2023; Laurençon et al., 2023), OPT, Pythia (Biderman et al., 2023) and Phi (Abdin et al., 2024) families. ‡ marks the ten evaluated downstream. Bits per character are omitted because within a script they are a fixed multiple of bits per byte, $2.651\times$ for native Sinhala and $1.003\times$ for its romanization. Note that the PPL and BPB orderings disagree almost completely.